

Facts: A manufacturing cycle efficiency of 40% means that

- Value-added activities are being performed 40% of the time.
- The typical order is being worked on 40% of the time.

$$\begin{array}{rcl}
 \text{AOA} & \text{NOI} & \text{NOI / AOA} \\
 200,000 & 24,000 & = .12 \text{ or } 12\% \\
 & & 24,000 / 200,000 \\
 \text{NOI} & - & (\text{AOA} \times \text{min Rate}) \\
 24,000 & - & (\$200,000 \times 10\%) = \\
 24,000 & - & 20,000 \\
 & = & \boxed{4,000}
 \end{array}$$

* When a manager is evaluated on residual income, an investment is acceptable when net operating income for the investment is above the minimum required return on average operating assets

* A manager might reject a proposal using ROI that the manager would accept using residual income.

* When evaluating investment center, it is probably better to focus on the percentage change in residual income from year to year rather than on the absolute amount of the residual income

$$\begin{array}{rcl}
 \text{NOI} & - & (\text{AOA} \times \text{min. rate return}) = \\
 42,000 & - & (300,000 \times 12\%) = \\
 42,000 & - & 36,000 = 6,000 \text{ Residual Income}
 \end{array}$$

* Performance measures should be tailored to the specific strategy of a company

If a balanced scorecard is correctly constructed, the performance measures should be linked together on a cause-and-effect basis.

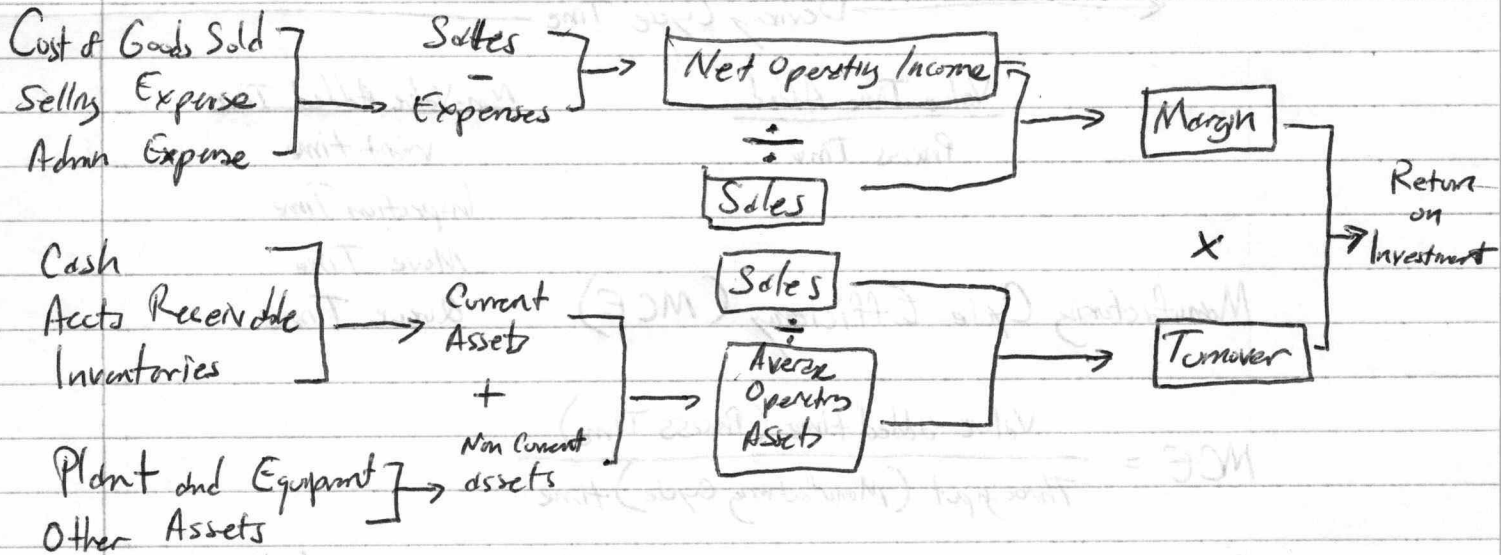
$$ROI = \text{Margin} \times \text{Turnover}$$

$$ROI = \frac{\text{Net operating Income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Average Operating Assets}}$$

$$= \frac{10,000}{100,000} \times \frac{100,000}{50,000}$$

$$ROI = 10\% \times 2 = 20\%$$

Elements of ROI



Residual income is the net operating income that an investment center earns above the minimum required return on its operating assets.

equation:

$$\text{Residual Income} = \text{Net operating Income} - \left(\text{Average operating Assets} \times \text{Minimum Required rate of return} \right)$$

Non-Financial performance measures: what drives organizational performance

Throughput (Manufacturing Cycle) time = Process time + Inspection time + Move time + Queue time
 Process time: amount of time work is actually being done on the product

Inspection time: the amount of time spent during

Move time: time required to move materials or partially completed products from workstation to workstation

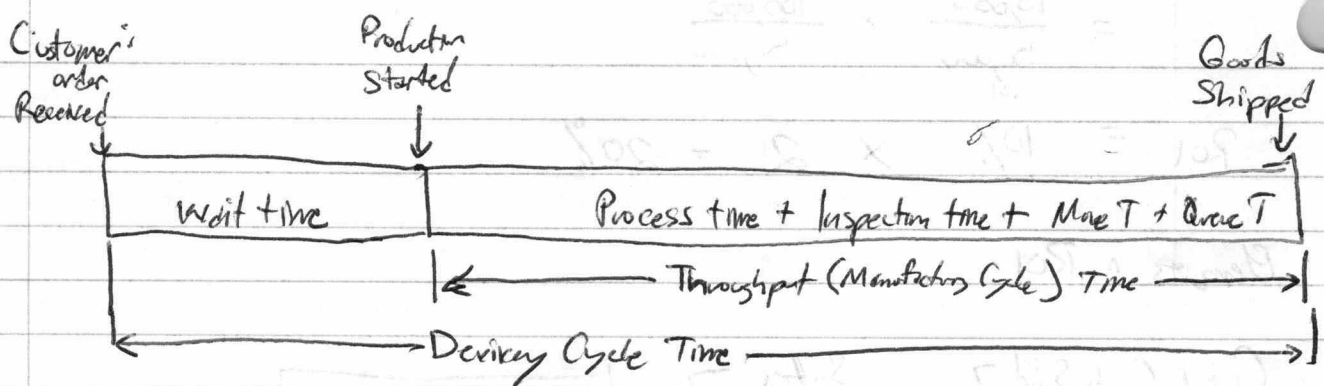
Queue time: the amount of time product waits to be worked on, moved, inspected, or shipped.

Only PROCESS TIME adds value to the product

Delivery Cycle Time: the elapsed time from when a customer order is received until the finished goods are shipped. The goal is to reduce this measure

Formula
Equation:

$$\text{Delivery cycle time} = \text{Wait time} + \text{throughput time}$$



Value Time Added
Process Time

Non-Value Added Time

Wait time

Inspection Time

Move Time

Queue Time

Manufacturing Cycle Efficiency (MCE)

$$MCE = \frac{\text{Value-added time (Process Time)}}{\text{Throughput (Manufacturing Cycle) time}}$$

Any non-value time results in an MCE of less than 1.

An MCE of .5 would mean that half of the total production time consists of inspection, moving, and similar non-value-added activities.

In many manufacturing companies the MCE is less than .01 (1%), which means 99% of the time, a unit in process is spent on activities that do not add value to the product.

Monitoring MCE help companies to reduce non-value-added activities and thus get products into the hands of customers more quickly and at a ~~lower~~ lower cost.

Solution:

$$\begin{aligned} 1) \quad \text{Throughput time} &= \text{Process time} + \text{Inspection time} + \text{Move T} + \text{Queue Time} \\ &= 2.0 \text{ days} + 0.4 \text{ days} + 0.6 \text{ days} + 5.0 \text{ days} \\ &= 8.0 \text{ days} \end{aligned}$$

2) Only process time represents value-added time; therefore, MCE would be computed as follows:

$$MCE = \frac{\text{Value-added time}}{\text{Throughput time}} = \frac{2.0 \text{ days}}{8.0 \text{ days}}$$

$$= 0.25$$

3) Thus, once put into production, a typical order is actually being worked on only 25% of the time.

Because the MCE is 25%, 75% (100% - 25%) of total production time is spent in non-value-added activities

$$\begin{aligned} 4) \text{ Delivery cycle time} &= \text{Wait time} + \text{Throughput time} \\ &= 17.0 \text{ days} + 8.0 \text{ days} \\ &= 25.0 \text{ days.} \end{aligned}$$

Balanced Scorecard: an integrated set of performance measures that are derived from ^{the} support the organization's strategy

RECAP: Throughput time = Process T + Inspection T + Move T + Queue T

RECAP: The higher a business segment's return on investment (ROI), the greater the profit earned per dollar invested in the segment's average operating assets.

RECAP: Net book value = acquisition cost less accumulated depreciation

Alternative to net book value is the gross cost of the asset, which ignores accumulated depreciation.

Using the Balanced Scorecard, ^{translates} top managers strategy into performance measures that employees can understand and influence.

Financial Measures: are lag indicators that report on the results of past actions.
top managers vs. lower level managers are ordinarily responsible for financial performance measures

A

B

C

Sales	9,000,000	2,000,000	4,500,000
NOI	540,000	280,000	
AOA	3,000,000	2,000,000	1,800,000
ROI	18%	14%	
Min Rate of return			
%	16%	16%	15%
\$	480,000	320,000	270,000
Residual Income	60,000	40,000	90,000

(A)

1) To get net operating income of Company A = AOA x ROI

$$\begin{array}{c} \text{AOA} \\ 3,000,000 \end{array} \times \begin{array}{c} \text{ROI} \\ .18 \end{array} = \boxed{540,000} \text{ NOI}$$

2) To get min required rate of return of company A = AOA x min req rate of return

$$\begin{array}{c} \text{AOA} \\ 3,000,000 \end{array} \times .16 = \boxed{480,000}$$

3) To get residual income of company A = NOI - min required rate of return

$$\begin{array}{c} \text{NOI} \\ 540,000 \end{array} - \begin{array}{c} \text{Min ROR} \\ 480,000 \end{array} = \boxed{60,000} \text{ Residual Income}$$

(B)

1) To find AOA for company B = NOI divided by ROI % =

$$\begin{array}{c} \text{NOI} \\ 280,000 \end{array} \div \begin{array}{c} \text{ROI \%} \\ .12 \end{array} = 2,000,000$$

2) Min Required Rate of Return for Co. B = MRROR / AOA

$$\begin{array}{c} \text{MRROR} \\ 320,000 \end{array} \div 2,000,000 = .16$$

3) Residual income of Co. B = NOI - MRROR

$$\begin{array}{c} \text{NOI} \\ 280,000 \end{array} - \begin{array}{c} \text{MRROR} \\ 320,000 \end{array}$$

Company C.

2008

1) $\text{Company C ROI\%} = \text{NOI} / \text{AOA}$

$\text{NOI} \div \text{AOA} = \text{ROI\%}$

$360,000 \div 1,800,000 = .2 \text{ or } 20\%$

$\text{MMROR\$} = \text{AOA} \times \text{MMROR\%}$

2) $\text{AOA} \times \text{MMROR}$

$1,800,000 \times .15 = \boxed{270,000} \text{ MMROR\$}$

3) $\text{Net operating income of Co. C} = \text{MMROR\$} + \text{Residual income}$

$\text{MMROR\$} + \text{Residual} = 360,000$
 $270,000 + 90,000 = 360,000$

$\text{ROI} = \text{Margin}$

$\text{Margin} \frac{\text{NOI}}{\text{Sales}} \times \text{Turnover} \frac{\text{Sales}}{\text{AOA}}$

$\text{Margin} \frac{150,000}{3,000,000} \times \frac{3,000,000}{750,000}$
 $.05 \times 4.0 = .2 (20\%)$

Sales 3,000,000

NOI 4,500,000

AOA 750,000

300,000

$\text{margin} \frac{4,500,000}{3,000,000} \times \text{Turnover} \frac{3,000,000}{750,000}$

1.5

x

4

=

6.0

$$NOI \times \frac{.5}{1}$$

divide by AOA

$$150,000 \times .5 = 75,000$$

$$75,000 \div 750,000 = 1.0$$

Sales 4,500,000

Sales 4,000,000

NOI 350,000

AOA 950,000

$$ROI = \frac{NOI}{Sales} \times \frac{Sales}{AOA}$$

$$ROI = \frac{350,000}{4,000,000} \times \frac{4,000,000}{950,000}$$

$$.0875 \times 4.210 = .3684 \text{ or } 37\%$$